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N + - Network Fundamentals & Building Networks.

Network Cable and Connectors.

TASK 1.

- ❖ Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

➤ Cat6 Ethernet Cable: T568B Standard

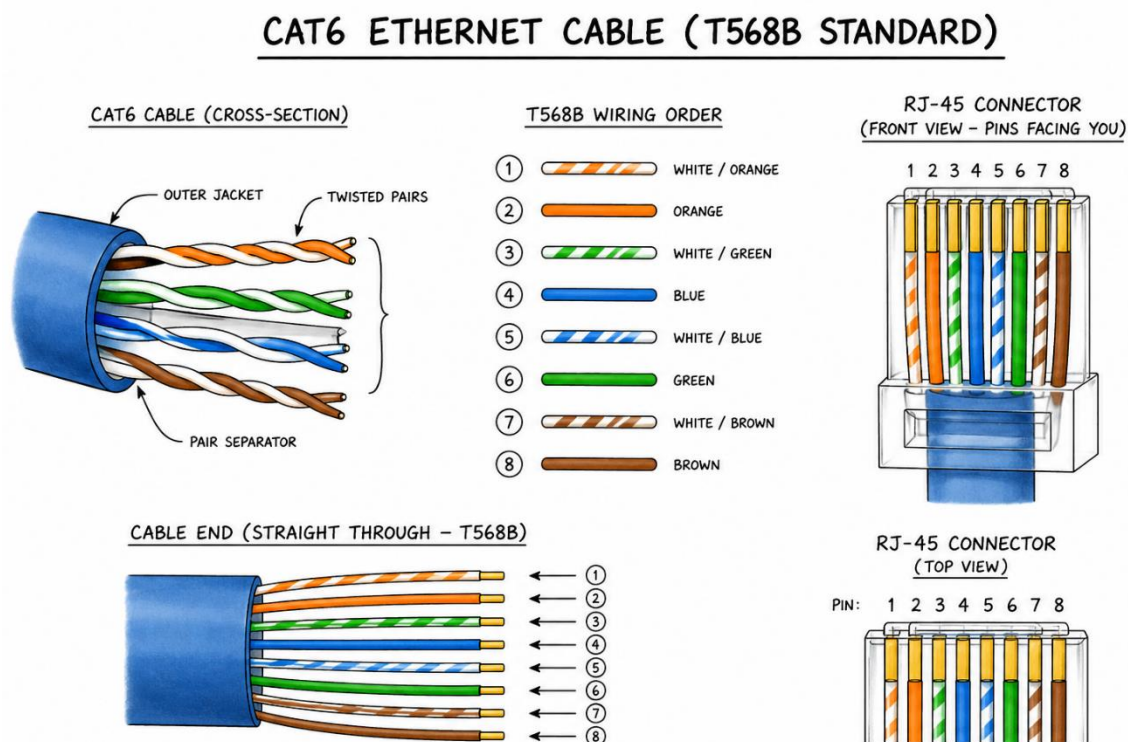


Figure : Cat6 Ethernet Cable with T568B Wiring Order and RJ-45 Pin Configuration.

Explanation :

- ✓ A Cat6 Ethernet cable contains four twisted pairs, consisting of eight wires. The T568B standard defines the specific colour sequence of these wires from Pin 1 to Pin 8 of the RJ-45 connector.

The T568B standard specifies the following colour sequence for the eight wires of a Cat6 Ethernet cable:

Pin	Wire Colour
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

TASK 2.

❖ List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

➤ Types of Network Cables and Their Real-World Uses:

Cable Type	Application / Device	Why It Is Used
Cat5e	Wi-Fi Router	Provides a stable wired connection with sufficient speed for normal internet usage.
Cat6	Zomato POS Terminal	Provides faster and more reliable data transfer, helping orders and payments process smoothly.
Fiber Optic	IPL Streaming Server	Supports very high-speed data transmission over long distances with minimal signal loss, making it ideal for live streaming.

Conclusion:

Different network cables are chosen according to the **speed, distance, and reliability** required. Cat5e is suitable for regular networking, Cat6 provides better performance for devices such as POS terminals, while fiber optic is preferred for **high-speed and long-distance communication**, such as large-scale video streaming.

TASK 3.

❖ **Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.**

➤ **Crimping an RJ-45 Connector:**

➤ **Step by Step Process:**

✂ **Cut the Cable:**

Take the required length of Ethernet cable and cut the end neatly using a cable cutter.

✂ **Remove the Outer Cover:**

Carefully strip a small portion of the outer jacket to expose the eight twisted wires inside.

✂ **Arrange the Wires:**

Untwist and straighten all eight wires. Arrange them according to the **T568B color sequence**.

✂ **Trim the Wires:**

Make the ends of all eight wires equal in length so that they can easily fit inside the RJ-45 connector.

✂ **Insert the Wires:**

Insert all eight wires into the RJ-45 connector. Make sure each wire reaches the front of the connector and stays in the correct order.

✂ **Crimp the Connector:**

Place the RJ-45 connector into the crimping tool and press it firmly. This locks the connector onto the cable.

⌘ **Check the Connection:**

Finally, check that the wires are still in the correct order and the connector is firmly attached. A cable tester can also be used to confirm that the connection works properly.

🔗 **Two Common Mistkes to Avoid:**

⌘ **Wrong wire sequence:** Arranging the wires in the wrong T568B order can cause the cable to work incorrectly or not work at all.

⌘ **Loose or incomplete insertion:** If the wires do not reach the end of the RJ-45 connector before crimping, the connection may become loose or fail.

🏁 **Conclusion:**

Crimping an RJ-45 connector requires **correct wire arrangement, proper insertion, and firm crimping**. Taking care at each step helps create a reliable Ethernet connection.

TASK 4.

❖ **Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.**

➤ **For an IPL screening gaming zone, the following cables and connectors would be suitable:**

Connection	Cable & Connector	Reason
Gaming Console → Screen	HDMI Cable + HDMI Connector	Provides high-quality video and audio with low delay, making it suitable for gaming.
Screens/Consoles → Network Switch	Cat6 Cable + RJ-45 Connector	Provides fast and stable network connectivity for multiple devices.
Main Network → Server	Fiber Optic Cable + Fiber Connector	Supports very high-speed data transfer over long distances with low signal loss.

 Conclusion:

Using **HDMI, Cat6, and fiber optic cables** ensures high-speed, reliable, and smooth connections for both gaming and IPL screening.

TASK 5.

- ❖ Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects, Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



Figure : Cat6 Ethernet Cable with RJ-45 Connector in Use.

⌘ **Type:** Cat6 Ethernet Cable with RJ-45 Connector.

⌘ **Devices Connected:** Network Switch and Office Computers.

 Description:

The blue Cat6 cables are connected to a network switch using RJ-45 connectors. They provide a **high-speed and reliable wired connection** between the computers and the network, ensuring efficient data transmission.